3.4 Graph of Tangent

Recall: 3.4 Graphs of Secant and Cosecant

Name _____ **Solo – 9 minutes**

1. Use the guide function $\cos x$ for $y = \sec x$ on $0^\circ \leqslant x \leqslant 360^\circ$. Where are the vertical asymptotes (VA)? At which x do the two graphs match?

2. Use the guide function $\sin x$ for $y = \csc x$ on $0^\circ \leqslant x \leqslant 360^\circ$. Where are the VA? At which x do the two graphs match?

3. Fill in the table.

x	-45°	0°	45°	90°	135°
$\sin x$					
$\cos x$					
$\dfrac{\sin x}{\cos x}$					

Treat $y = \dfrac{\sin x}{\cos x}$ as a rational function. Where would its zeros and VA be?